# SEMICONDUCTOR MISSION AND ELECTRONICS MANUFACTURING

DESIGN-TO-DEVICE RAIL → SEMICONDUCTOR BUSINESS-MODEL MATRIX → FRONT-END VERSUS BACK-END → NODE AND LITHOGRAPHY LADDER → ADVANCED INTEGRATION MAP → MATERIALS TAXONOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

## STAGE 00 DESIGN-TO-DEVICE RAIL

DESIGN-TO-DEVICE RAIL | ARCHITECTURE, VERIFICATION AND TAPE-OUT | TRANSISTOR AND INTERCONNECT LAYERS ON WAFER | WAFER BECOMES INDIVIDUAL DIES | ASSEMBLE, TEST, MARK AND PACKAGE | DOWNSTREAM ELECTRONICS INTEGRATION

DESIGN / IP / EDA → architecture, verification and tape-out → FOUNDRY / FAB → transistor and interconnect layers on wafer → DICING → wafer becomes individual dies → ATMP / OSAT → assemble, test, mark and package → BOARD / MODULE / DEVICE → downstream electronics integration

### STARTING CONDITION / INPUT

- DESIGN / IP / EDA → architecture, verification and tape-out
- FOUNDRY / FAB → transistor and interconnect layers on wafer

### SCIENTIFIC OR ENGINEERING MECHANISM

- DICING → wafer becomes individual dies
- ATMP / OSAT → assemble, test, mark and package

### OUTPUT / FAILURE BOUNDARY

- BOARD / MODULE / DEVICE → downstream electronics integration
- DESIGN / IP / EDA → architecture, verification and tape-out

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOARD / MODULE / DEVICE → downstream electronics integration.

01

## STAGE 01 SEMICONDUCTOR BUSINESS-MODEL MATRIX

SEMICONDUCTOR BUSINESS-MODEL MATRIX | PHYSICAL WAFER-MANUFACTURING PLANT | DESIGNS AND OUTSOURCES WAFER MANUFACTURE | MANUFACTURES TO CUSTOMERS' DESIGNS | DESIGNS AND FABRICATES ITS OWN PRODUCTS | PLANT TYPE AND FIRM MODEL ANSWER DIFFERENT QUESTIONS

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| FAB → physical wafer-manufacturing plant | FOUNDRY → manufactures to customers' designs | RULE → plant type and firm model answer different questions |
| FABLESS → designs and outsources wafer manufacture | IDM → designs and fabricates its own products | FAB → physical wafer-manufacturing plant |

### SYSTEM / DEVICE / INSTITUTION

- FAB → physical wafer-manufacturing plant
- FABLESS → designs and outsources wafer manufacture

### DECISIVE TECHNICAL DISTINCTION

- FOUNDRY → manufactures to customers' designs
- IDM → designs and fabricates its own products

### STATUS / UNIT / EVIDENCE LIMIT

- RULE → plant type and firm model answer different questions
- FAB → physical wafer-manufacturing plant

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → plant type and firm model answer different questions.

02

## STAGE 02 FRONT-END VERSUS BACK-END

FRONT-END VERSUS BACK-END | FRONT END | DEPOSITION, LITHOGRAPHY, ETCHING, DOPING, METALLISATION | PROCESSED WAFER CONTAINING DIES | BACK END | BONDING, ENCAPSULATION, TESTING, MARKING, PACKAGING | PROCESS DESCRIPTION; OSAT | OUTSOURCED SERVICE MODEL

| SYSTEM / DEVICE / INSTITUTION | DECISIVE TECHNICAL DISTINCTION | STATUS / UNIT / EVIDENCE LIMIT |
|---|---|---|
| FRONT END → deposition, lithography, etching, doping, metallisation | BACK END → bonding, encapsulation, testing, marking, packaging | TRAP → back-end capacity is not front-end fabrication |
| OUTPUT → processed wafer containing dies | ATMP → process description; OSAT → outsourced service model | FRONT END → deposition, lithography, etching, doping, metallisation |

### SYSTEM / DEVICE / INSTITUTION

- FRONT END → deposition, lithography, etching, doping, metallisation
- OUTPUT → processed wafer containing dies

### DECISIVE TECHNICAL DISTINCTION

- BACK END → bonding, encapsulation, testing, marking, packaging
- ATMP → process description; OSAT → outsourced service model

### STATUS / UNIT / EVIDENCE LIMIT

- TRAP → back-end capacity is not front-end fabrication
- FRONT END → deposition, lithography, etching, doping, metallisation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → back-end capacity is not front-end fabrication.

03

## STAGE 03 NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER | BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES | MATURE AND MID-RANGE LITHOGRAPHY ROUTE | LEADING EDGE | RISING COMPLEXITY AND CAPITAL INTENSITY | CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT | VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

MATURE / MID-RANGE → broad automotive, industrial and power uses

DUV → mature and mid-range lithography route → LEADING EDGE → rising complexity and capital intensity → EUV → concentrated equipment and export-control chokepoint

## STAGE 03 NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER · BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES · MATURE AND MID-RANGE LITHOGRAPHY ROUTE · LEADING EDGE · RISING COMPLEXITY AND CAPITAL INTENSITY · CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT · VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

MATURE / MID-RANGE → broad automotive, industrial and power uses

DUV → mature and mid-range lithography route → LEADING EDGE → rising complexity and capital intensity → EUV → concentrated equipment and export-control chokepoint

**STARTING CONDITION / INPUT**
- MATURE / MID-RANGE → broad automotive, industrial and power uses
- DUV → mature and mid-range lithography route

**SCIENTIFIC OR ENGINEERING MECHANISM**
- LEADING EDGE → rising complexity and capital intensity
- EUV → concentrated equipment and export-control chokepoint

**OUTPUT / FAILURE BOUNDARY**
- RULE → verify every project-specific node claim officially
- MATURE / MID-RANGE → broad automotive, industrial and power uses

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → verify every project-specific node claim officially.**

## STAGE 04 ADVANCED INTEGRATION MAP

ADVANCED INTEGRATION MAP · SPECIALISED DIES COMBINED AS ONE SYSTEM · DENSER PACKAGE-LEVEL INTEGRATION · THROUGH-SILICON VIAS · VERTICAL INTERCONNECT ROUTE · HETEROGENEOUS INTEGRATION · COMBINE UNLIKE FUNCTIONS OR PROCESSES · PERFORMANCE GAINS NEED NOT BEGIN WITH SMALLER LITHOGRAPHY

CHIPLETS → specialised dies combined as one system

2.5D / 3D → denser package-level integration → THROUGH-SILICON VIAS → vertical interconnect route → HETEROGENEOUS INTEGRATION → combine unlike functions or processes

**CONCEPT / ARCHITECTURE**
- CHIPLETS → specialised dies combined as one system
- 2.5D / 3D → denser package-level integration

**MECHANISM / APPLICATION**
- THROUGH-SILICON VIAS → vertical interconnect route
- HETEROGENEOUS INTEGRATION → combine unlike functions or processes

**READINESS / STATUS LIMIT**
- INSIGHT → performance gains need not begin with smaller lithography
- CHIPLETS → specialised dies combined as one system

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INSIGHT → performance gains need not begin with smaller lithography.**

## STAGE 05 MATERIALS TAXONOMY

MATERIALS TAXONOMY · COMPOUND SEMICONDUCTORS · APPLICATION FAMILIES · SILICON PHOTONICS · SILICON OPTICAL-INTERCONNECT PLATFORM · POLICY GROUPING · MAY PLACE UNLIKE TECHNOLOGIES TOGETHER · SCHEME NOMENCLATURE DOES NOT CHANGE MATERIAL PHYSICS

COMPOUND SEMICONDUCTORS → GaN / GaAs / InP / SiC

POWER / RF / OPTOELECTRONICS → application families → SILICON PHOTONICS → silicon optical-interconnect platform → POLICY GROUPING → may place unlike technologies together

**CONCEPT / ARCHITECTURE**
- COMPOUND SEMICONDUCTORS → GaN / GaAs / InP / SiC
- POWER / RF / OPTOELECTRONICS → application families

**MECHANISM / APPLICATION**
- SILICON PHOTONICS → silicon optical-interconnect platform
- POLICY GROUPING → may place unlike technologies together

**READINESS / STATUS LIMIT**
- RULE → scheme nomenclature does not change material physics
- COMPOUND SEMICONDUCTORS → GaN / GaAs / InP / SiC

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → scheme nomenclature does not change material physics.**

## STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN · RELIABLE POWER + ULTRA-PURE WATER · STABLE PROCESSING · GASES + CHEMICALS + EQUIPMENT · CONTROLLED PROCESS STEPS · REPEATED RUNS · YIELD LEARNING · RELIABILITY EVIDENCE

RELIABLE POWER + ULTRA-PURE WATER → stable processing

GASES + CHEMICALS + EQUIPMENT → controlled process steps → REPEATED RUNS → yield learning → RELIABILITY EVIDENCE → customer qualification

**STARTING CONDITION / INPUT**
- RELIABLE POWER + ULTRA-PURE WATER → stable processing

**SCIENTIFIC OR ENGINEERING MECHANISM**
- REPEATED RUNS → yield learning

**OUTPUT / FAILURE BOUNDARY**
- OUTCOME → inauguration alone is not competitive production

06

## STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN · RELIABLE POWER + ULTRA-PURE WATER · STABLE PROCESSING · GASES + CHEMICALS + EQUIPMENT · CONTROLLED PROCESS STEPS · REPEATED RUNS · YIELD LEARNING · RELIABILITY EVIDENCE

RELIABLE POWER + ULTRA-PURE WATER → stable processing

GASES + CHEMICALS + EQUIPMENT → controlled process steps → REPEATED RUNS → yield learning → RELIABILITY EVIDENCE → customer qualification

### STARTING CONDITION / INPUT
- RELIABLE POWER + ULTRA-PURE WATER → stable processing
- GASES + CHEMICALS + EQUIPMENT → controlled process steps

### SCIENTIFIC OR ENGINEERING MECHANISM
- REPEATED RUNS → yield learning
- RELIABILITY EVIDENCE → customer qualification

### OUTPUT / FAILURE BOUNDARY
- OUTCOME → inauguration alone is not competitive production
- RELIABLE POWER + ULTRA-PURE WATER → stable processing

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → inauguration alone is not competitive production.**

07

## STAGE 07 INSTITUTION AND SCHEME MAP

INSTITUTION AND SCHEME MAP · PARENT MINISTRY · IMPLEMENTING AGENCY · SEMICON INDIA · SCHEME PACKAGE · DESIGN-FOCUSED CHANNEL · SEPARATE COMPLEMENTARY ELECTRONICS INCENTIVES

MeitY → parent ministry

ISM → implementing agency → SEMICON INDIA → scheme package → DLI → design-focused channel

### CONCEPT / ARCHITECTURE
- MeitY → parent ministry
- ISM → implementing agency

### MECHANISM / APPLICATION
- SEMICON INDIA → scheme package
- DLI → design-focused channel

### READINESS / STATUS LIMIT
- SPECS / PLI → separate complementary electronics incentives
- MeitY → parent ministry

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECS / PLI → separate complementary electronics incentives.**

08

## STAGE 08 SEMICON 1.0 TO 2.0

SEMICON 1.0 TO 2.0 · SEMICON INDIA 1.0 · RS 76,000 CRORE UMBRELLA · FOUR CHANNELS · FABS, DISPLAYS, COMPOUND/SENSOR/ATMP, DLI · 15 JUL 2026 · SEMICON 2.0 CABINET APPROVAL · SEMICON 2.0

SEMICON INDIA 1.0 → Rs 76,000 crore umbrella → FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI → 15 JUL 2026 → Semicon 2.0 Cabinet approval → SEMICON 2.0 → Rs 1,27,500 crore approved outlay → SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent

### CONCEPT / ARCHITECTURE
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella
- FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

### MECHANISM / APPLICATION
- 15 JUL 2026 → Semicon 2.0 Cabinet approval
- SEMICON 2.0 → Rs 1,27,500 crore approved outlay

### READINESS / STATUS LIMIT
- SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent.**

09

## STAGE 09 FACILITY EVIDENCE MAP

FACILITY EVIDENCE MAP · FAB; LATER NAMED COMPOUND-FAB PLUS ATMP APPROVAL · OFFICIALLY IDENTIFIED OSAT · ATMP LOCATION · OFFICIALLY IDENTIFIED BACK-END CLUSTER AND INAUGURATIONS · SUCHI SEMICON OSAT APPROVAL · LOCATION MUST TRAVEL WITH PROJECT AND STATUS VERB · MUST REMEMBER

DHOLERA → fab; later named compound-fab plus ATMP approval

MORIGAON → officially identified OSAT / ATMP location → SANAND → officially identified back-end cluster and inaugurations → SURAT → Suchi Semicon OSAT approval

### CONCEPT / ARCHITECTURE
- DHOLERA → fab; later named compound-fab plus ATMP approval
- MORIGAON → officially identified OSAT / ATMP location

### MECHANISM / APPLICATION
- SANAND → officially identified back-end cluster and inaugurations
- SURAT → Suchi Semicon OSAT approval

### READINESS / STATUS LIMIT
- RULE → location must travel with project and status verb
- MUST REMEMBER: The semiconductor chain separates design/IP, wafer fabrication, compound...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location must travel with project and status verb.**

DHOLERA → fab; later named compound-fab plus ATMP approval

MORIGAON → officially identified OSAT / ATMP location → SANAND → officially identified back-end cluster and inaugurations → SURAT → Suchi Semicon OSAT approval

**CONCEPT / ARCHITECTURE**
- DHOLERA → fab; later named compound-fab plus ATMP approval
- MORIGAON → officially identified OSAT / ATMP location

**MECHANISM / APPLICATION**
- SANAND → officially identified back-end cluster and inaugurations
- SURAT → Suchi Semicon OSAT approval

**READINESS / STATUS LIMIT**
- RULE → location must travel with project and status verb
- MUST REMEMBER: The semiconductor chain separates design/IP, wafer fabrication, compound...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location must travel with project and status verb.**

10

## STAGE 10 STATUS FIREWALL

STATUS FIREWALL | POLICY PERMISSION AND SUPPORT DECISION | FINANCING OR CONSTRUCTION MILESTONE | PHYSICAL OR DEMONSTRATION MILESTONE | QUALIFIED COMMERCIAL SUPPLY | SEPARATE OPERATING EVIDENCE | NEVER SKIP A RUNG OR INVENT PRODUCTION | CLOSE DISTINCTION

APPROVED → policy permission and support decision

AGREEMENT / GROUNDBREAKING → financing or construction milestone → INAUGURATED / PRESENTED → physical or demonstration milestone → QUALIFIED COMMERCIAL SUPPLY → separate operating evidence

**SYSTEM / DEVICE / INSTITUTION**
- APPROVED → policy permission and support decision
- AGREEMENT / GROUNDBREAKING → financing or construction milestone

**DECISIVE TECHNICAL DISTINCTION**
- INAUGURATED / PRESENTED → physical or demonstration milestone
- QUALIFIED COMMERCIAL SUPPLY → separate operating evidence

**STATUS / UNIT / EVIDENCE LIMIT**
- RULE → never skip a rung or invent production
- CLOSE DISTINCTION: Design is not fabrication, fab is not ATMP, packaging is not mere...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never skip a rung or invent production.**

11

## STAGE 11 PYQ ANSWER SPINE

PYQ ANSWER SPINE | VALUE CHAIN, FAB MODELS AND ATMP | ISM, FOUR SEMICON 1.0 CHANNELS AND SEMICON 2.0 PILLARS | TOOLS, UTILITIES, TALENT, YIELD AND QUALIFICATION | DHRUV64 AND FACILITIES WITH EXACT STATUS VERBS | NO PROJECT TOTAL, NODE, COMMERCIAL OUTPUT OR ANSWER KEY | EVIDENCE LIMIT | DATE PROJECT CLAIMS; STATE VALUE-CHAIN

DEFINE → value chain, fab models and ATMP / OSAT → MAP → ISM, four Semicon 1.0 channels and Semicon 2.0 pillars → ANALYSE → tools, utilities, talent, yield and qualification → EVIDENCE → DHRUV64 and facilities with exact status verbs → QUALIFY → no project total, node, commercial output or answer key

**CONCEPT / ARCHITECTURE**
- DEFINE → value chain, fab models and ATMP / OSAT
- MAP → ISM, four Semicon 1.0 channels and Semicon 2.0 pillars

**MECHANISM / APPLICATION**
- ANALYSE → tools, utilities, talent, yield and qualification
- EVIDENCE → DHRUV64 and facilities with exact status verbs

**READINESS / STATUS LIMIT**
- QUALIFY → no project total, node, commercial output or answer key
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date project claims; state value-chain...

**MECHANISM / VERDICT —** EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Date project claims; state value-chain...

**ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ANALYSE → tools, utilities, talent, yield and qualification.**

E

## EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | PACKAGING-FIRST VERSUS FAB-FIRST DEBATE | INDIA MAY OBTAIN FASTER LEARNING, CUSTOMER RELATIONSHIPS AND | EXPORT INTEGRATION THROUGH ATMP/OSAT AND COMPOUND-SEMICONDUCTOR NICHES BEFORE MASTERING | SUBSIDY VERSUS ECOSYSTEM DEBATE | GENEROUS FISCAL SUPPORT CAN ATTRACT PROPOSALS, BUT | PROCESS ENGINEERS, UTILITIES, MATERIALS AND ANCHOR CUSTOMERS DETERMINE WHETHER | STRATEGIC AUTONOMY VERSUS COST EFFICIENCY

**OPTIONAL SCIENTIFIC / ENGINEERING DEPTH**
- CAUTION: Packaging-first versus fab-first debate: India may obtain faster learning, customer relationships and
- export integration through ATMP/OSAT and compound-semiconductor niches before mastering complex leading-edge fabrication.
- CAUTION: Subsidy versus ecosystem debate: generous fiscal support can attract proposals, but
- process engineers, utilities, materials and anchor customers determine whether plants remain viable.

**READINESS, UNIT OR STATUS LIMIT**
- CAUTION: Strategic autonomy versus cost efficiency: semiconductors are treated as strategic infrastructure, yet full autarky is unrealistic
- the more realistic goal is selective resilience in critical layers.
- CAUTION: Technology-neutral versus technology-specific support: India must decide whether to prioritize broad ecosystem depth or
- a few strategically valuable segments such as power electronics, sensors, compound semiconductors and packaging.

**COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE**
- CAUTION: Semiconductors increasingly shape AI, telecom, defence electronics, automobiles, power systems and industrial automation.
- CAUTION: Domestic packaging and design capability can reduce vulnerability even when India still imports large volumes of wafers or equipment.
- CAUTION: Semiconductor success improves the quality of India's electronics manufacturing
- story by moving beyond screwdriver assembly toward deeper value capture.

**OPTIONAL STATUS —** This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels